

Smart Campus Digital Twin Project (BI + Software Engineering Guide)

1. Project Overview

This project simulates a real-time digital twin of a campus with BI analytics, real-time tracking, and predictive insights.

2. Architecture

Components: Data Simulator, Backend API, Database (PostgreSQL), Real-Time Engine (WebSockets), Flutter Frontend, BI Layer.

3. Tech Stack

Frontend: Flutter Backend: FastAPI or Node.js Database: PostgreSQL Real-time: WebSockets BI: Power BI / Pandas

4. Development Steps

- Setup project and Git repository
- Design database schema (Star Schema)
- Build data simulator (Python)
- Develop backend API
- Implement real-time communication
- Build Flutter UI
- Integrate BI dashboards
- Testing & optimization
- Deployment

5. Scrum Plan

Sprint	Duration	Goal
Sprint 1	1 week	Setup + Design

Sprint 2	2 weeks	Backend + DB
Sprint 3	2 weeks	Flutter UI
Sprint 4	1 week	Real-time + BI
Sprint 5	1 week	Testing + Deploy

6. Estimated Timeline

Phase	Duration
Planning	1 week
Development	5 weeks
Testing	1 week
Deployment	1 week

7. UI Ideas

Dashboard, Live Map, Analytics Charts, Alerts Panel

8. Conclusion

This project combines BI, real-time systems, and mobile development into a strong portfolio project.